

Version 4 - Generated Data + SMOTE + Optuna - Feature Used / Dropped Audit

เทียบ source clean dataset กับ df_engineered และระบุว่า feature ใดใช้ train model / feature ใดตัดทิ้ง

Model Summary

Model	Dataset	df_engineered	Used Features	Accuracy	Recall	F1	AUC	Rating
XGBoost + SMOTE + Optuna	clean_dataset_v4_generated.csv	df_engineered_v4_generated.csv	180	83.45%	46.39%	45.69%	85.38%	C

Source Artifacts

artifact	path	columns	sample_rows_read
source_data	docs\version 4\data\processed\clean_dataset_v4_generated.csv	65	5
df_engineered	docs\version 4\data\features\df_engineered_v4_generated.csv	123	5
used_feature_list	docs\version 4\reports\model_evaluation\v4_used_features.csv	1	5

Used Feature Groups

group	feature_count	example_features
product	65	brand_AromaX, brand_BlendIt, brand_BoomBox, brand_CookEasy, brand_DenimCo
order_channel_payment	37	channel_orders_before, channel_returns_before, channel_type_Mobile_App, channel_type_Shopee, channel_type_TV_Show
other	20	cat_orders_before, cat_returns_before, courier_name_EcoDelivery, courier_name_FastShip, courier_name_SafeLogistics
customer_profile	14	age, age_group_25, age_group_26_35, age_group_36_45, age_group_46_55
geography	12	coverage_region_Bangkok_Only, coverage_region_Nationwide, province_Bangkok, province_Chiang_Mai, province_Chonburi
customer_history	11	brand_return_rate_pti, cat_return_rate_pti, channel_return_rate_pti, courier_return_rate_pti, cust_orders_before
price_discount	11	campaign_discount_pct, discount_amount_ratio, discount_applied_amount, is_high_discount, log_total_amount
time	10	avg_delivery_days, customer_age_days, customer_tenure_months, days_since_last_order, delivery_time_expected_days

Used Features (sample)

age, age_group_25, age_group_26_35, age_group_36_45, age_group_46_55, age_group_56_, avg_delivery_days, brand_AromaX, brand_BlendIt, brand_BoomBox, brand_CookEasy, brand_DenimCo, brand_GlowDrink, brand_GlowUp, brand_HomePro, brand_MuscleFuel, brand_OmegaPlus, brand_PureAir, brand_RunFast, brand_ShieldCase, brand_SilkTouch, brand_SkinLab, brand_SoundMax, brand_SunSafe, brand_TimeX, brand_UniStyle, brand_VitaBoost, brand_orders_before, brand_return_rate_pti, brand_returns_before, campaign_discount_pct, cat_orders_before, cat_return_rate_pti, cat_returns_before, category_Cosmetics, category_Electronics, category_Fashion, category_Home_Appliance, category_Supplement, category_channel_Cosmetics_Mobile_App, category_channel_Cosmetics_Shopee, category_channel_Cosmetics_TV_Show, category_channel_Cosmetics_TikTok, category_channel_Electronics_Mobile_App, category_channel_Electronics_Shopee, category_channel_Electronics_TV_Show, category_channel_Electronics_TikTok, category_channel_Fashion_Mobile_App, category_channel_Fashion_Shopee, category_channel_Fashion_TV_Show, category_channel_Fashion_TikTok, category_channel_Home_Appliance_Mobile_App, category_channel_Home_Appliance_Shopee, category_channel_Home_Appliance_TV_Show, category_channel_Home_Appliance_TikTok, category_channel_Supplement_Mobile_App, category_channel_Supplement_Shopee, category_channel_Supplement_TV_Show, category_channel_Supplement_TikTok, category_payment_Cosmetics_Bank_Transfer ...

Dropped / Not Used Feature Logic

- คัด leakage/post-event/target fields จำนวนประมาณ 16 feature เช่น delay_days, delivery_date, delivery_days, expected_delivery_date, is_returned, item_condition, refund_amount, return_date, return_id, return_reason
- คัด identifier/high-cardinality fields จำนวนประมาณ 13 feature เช่น courier_id, courier_name, customer_id, customer_name, customer_phone, order_id, product_id, product_name, promo_id, promo_name
- feature ที่เป็น raw date หรือ raw text จะถูกแปลงเป็น numeric/encoded/rolling feature ก่อน หรือไม่ใช่ถ้าไม่ช่วย performance
- feature ที่เพิ่มแล้ว performance ไม่ดีขึ้นหรือซ้ำซ้อนกับ feature สำคัญ จะถูกจัดเป็น NOT USED เพื่อลด resource/query cost

Feature Audit Detail (sample rows)

Feature	In Clean	In Engineered	Used	Status	Reason
age	True	False	True	USED	model input feature
age_group_25	False	True	True	USED	model input feature
age_group_26_35	False	True	True	USED	model input feature
age_group_36_45	False	True	True	USED	model input feature
age_group_46_55	False	True	True	USED	model input feature
age_group_56_	False	True	True	USED	model input feature
avg_delivery_days	True	False	True	USED	model input feature
brand_AromaX	False	True	True	USED	model input feature
brand_BlendIt	False	True	True	USED	model input feature
brand_BoomBox	False	True	True	USED	model input feature
brand_CookEasy	False	True	True	USED	model input feature
brand_DenimCo	False	True	True	USED	model input feature
brand	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
category	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
channel_type	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
courier_id	True	False	False	DROPPED	identifier or high-cardinality identity field
courier_name	True	False	False	DROPPED	identifier or high-cardinality identity field
courier_type	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
coverage_region	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
customer_id	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_name	True	False	False	DROPPED	identifier or high-cardinality identity field
customer_phone	True	False	False	DROPPED	identifier or high-cardinality identity field
delay_days	True	False	False	DROPPED	post-event/target/leakage field
delivery_date	True	False	False	DROPPED	post-event/target/leakage field
delivery_days	True	False	False	DROPPED	post-event/target/leakage field
ership_tier_Silver	False	True	False	NOT USED	engineered candidate not selected for this model
expected_delivery_date	True	False	False	DROPPED	post-event/target/leakage field
gender	True	False	False	DROPPED	raw field replaced by engineered/encoded feature or not required
is_returned	True	False	False	DROPPED	post-event/target/leakage field
item_condition	True	False	False	DROPPED	post-event/target/leakage field

Full CSV: docs\version 4\feature_documentation\v4_feature_used_unused_audit.csv